Introduction to multiple linear regression

Some statistics classes next semester

- STA 214 (Applied Generalized Linear Models)
  - Requires MTH 111 (Calc 1)
- STA 247 (Design and Sampling)
- STA 279 (Text Data)
- STA 310 (Probability)
  - Requires MTH 112 (Calc 2)
- Also consider MTH 121 (Linear algebra) for upper-level stats courses

Last time

$$\begin{aligned}\log(\text{Price per carat}) = \beta_0 + \beta_1 \text{ IsSI1} + \beta_2 \text{ IsSI2} + \beta_3 \text{ IsSI3} + \\ \beta_4 \text{ IsVS1} + \beta_5 \text{ IsVS2} + \beta_6 \text{ IsVVS1} + \\ \beta_7 \text{ IsVVS2} + \varepsilon\end{aligned}$$

> We want to test whether there is a relationship between Clarity and price per carat. What are our null and alternative hypotheses?

Last time

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_7 = 0$
H_A : at least one of $\beta_1, \beta_2, \ldots, \beta_7 \neq 0$

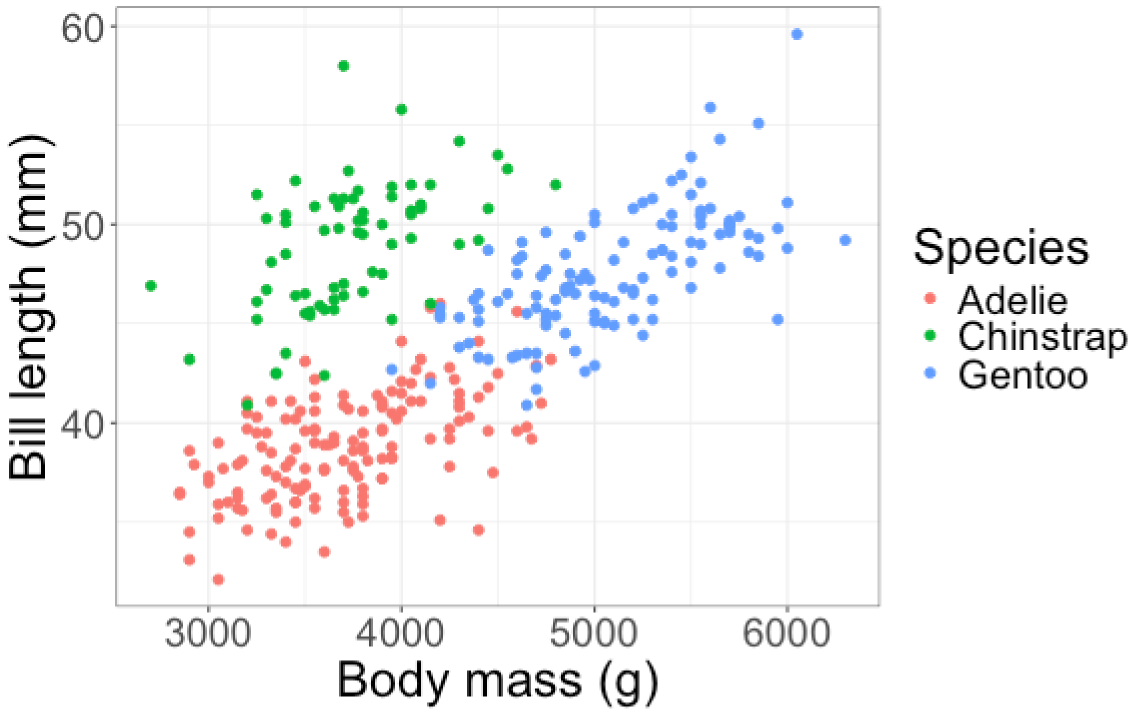
```
           Df Sum Sq Mean Sq F value  Pr(>F)
Clarity     7  3.038 0.43400  2.3832 0.02164 *
Residuals 343 62.463 0.18211
```

Last time

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.81518    0.10669  17.014   <2e-16 ***
ClaritySI1  -0.21875    0.12579  -1.739   0.0829 .
ClaritySI2  -0.24783    0.16714  -1.483   0.1391
ClaritySI3  -0.82695    0.32006  -2.584   0.0102 *
ClarityVS1  -0.01472    0.11347  -0.130   0.8969
ClarityVS2  -0.07700    0.11550  -0.667   0.5054
ClarityVVS1 -0.02433    0.13136  -0.185   0.8532
ClarityVVS2 -0.13451    0.12878  -1.044   0.2970
```

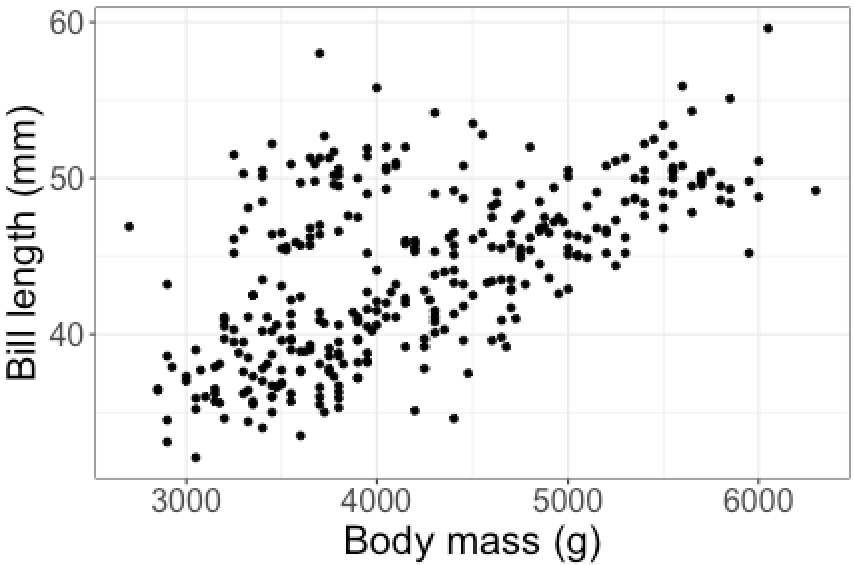
> Suppose that instead I just want to know whether there is a difference in price per carat between Internally Flawless (IF) diamonds and Very Slightly Included (VS1) diamonds. What would my hypotheses be for this test? How would I perform the test?

Relationships between > 2 variables



> How can we model the relationship between body mass, species, and bill length?

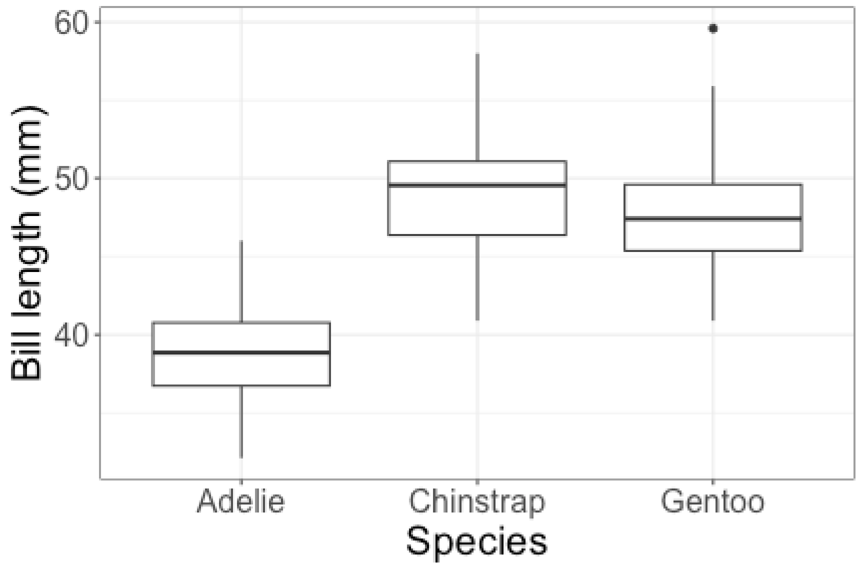
Regression with a single quantitative predictor



Model:

$$\text{bill length} = \beta_0 + \beta_1 \text{body mass} + \varepsilon$$

Regression with a single categorical predictor



Model:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

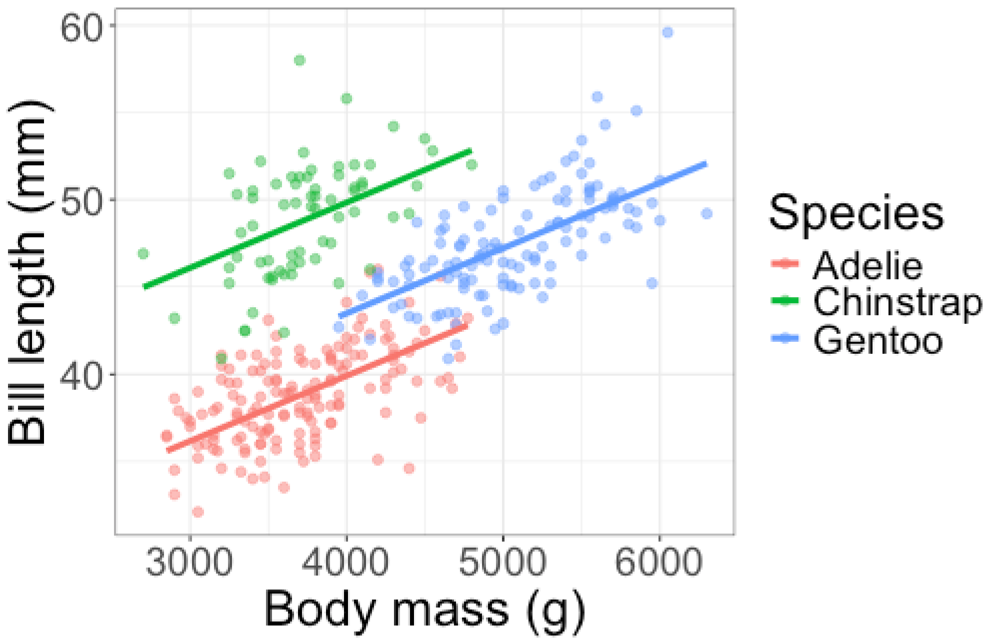
Regression with > 2 variables

$$\text{bill length} = \beta_0 + \beta_1 \text{body mass} + \varepsilon$$

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

> How can we include both species *and* body mass in the model?

Multiple regression



Model:

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ + \beta_3 \text{body mass} + \varepsilon$$

Multiple regression model

$$\begin{aligned}\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ + \beta_3 \text{body mass} + \varepsilon\end{aligned}$$

This gives us a separate regression line for each species of penguin.

> What is the regression line for Adelie penguins?

Multiple regression model

$$\begin{aligned}\text{bill length} &= \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ &+ \beta_3 \text{body mass} + \varepsilon\end{aligned}$$

This gives us a separate regression line for each species of penguin.

> What is the regression line for Adelie penguins?

$\text{bill length} = \beta_0 + \beta_3 \text{body mass} + \varepsilon$

> What about for Chinstrap and Gentoo penguins?

Multiple regression model

$$\begin{aligned}\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ + \beta_3 \text{body mass} + \varepsilon\end{aligned}$$

This gives us a separate regression line for each species of penguin:

$$\text{bill length} = \begin{cases} \beta_0 + \beta_3 \text{body mass} + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \beta_3 \text{body mass} + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \beta_3 \text{body mass} + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

Fitted model

$$\widehat{\text{bill length}} = 24.92 + 9.92\ \text{IsChinstrap} + 3.56\ \text{IsGentoo} \\ + 0.0037\ \text{body mass}$$

How would I interpret $\widehat{\beta}_0 = 24.92$?

Fitted model

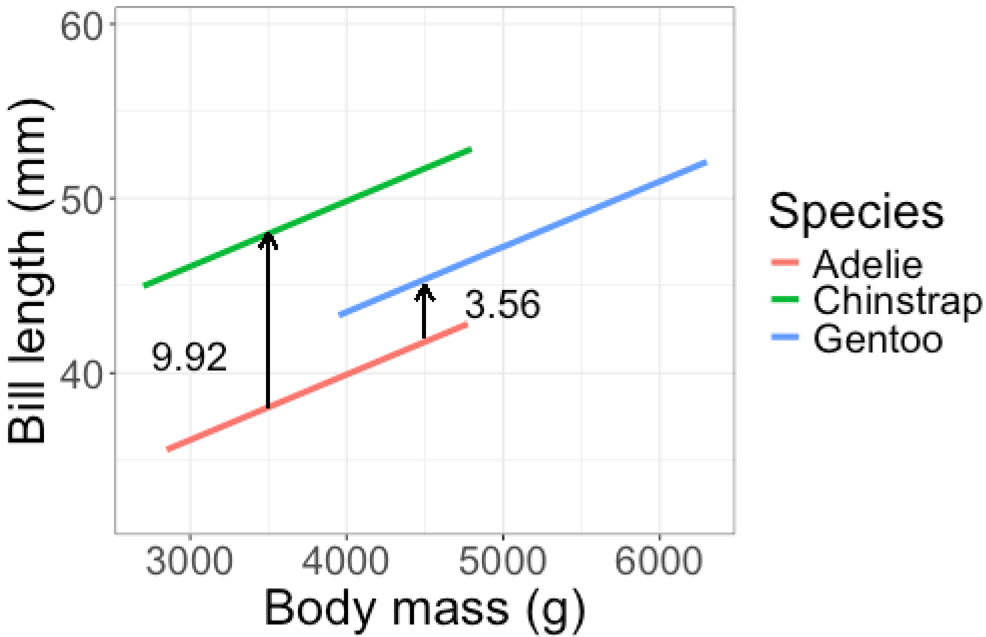
$$\widehat{\text{bill length}} = 24.92 + 9.92\ \text{IsChinstrap} + 3.56\ \text{IsGentoo} \\ + 0.0037\ \text{body mass}$$

> How would I interpret $\widehat{\beta}_0 = 24.92$?

The estimated bill length for an Adelie penguin weighing 0g. (not that meaningful)

> What about $\widehat{\beta}_1 = 9.92$ and $\widehat{\beta}_2 = 3.56$?

Interpreting regression coefficients



- $9.92 =$ estimated mean difference in bill length for a Chinstrap and Adelie penguin with the same body mass
- $3.56 =$ estimated mean difference in bill length for a Gentoo and Adelie penguin with the same body mass

Interpreting regression coefficients

$$\widehat{\text{bill length}} = 24.92 + 9.92\ \text{IsChinstrap} + 3.56\ \text{IsGentoo} + 0.0037\ \text{body mass}$$

> How would I interpret $\widehat{\beta}_3 = 0.0037$?

Interpreting regression coefficients

$$\begin{aligned}\widehat{\text{bill length}} = 24.92 + 9.92\ \text{IsChinstrap} + 3.56\ \text{IsGentoo} \\ + 0.0037\ \text{body mass}\end{aligned}$$

> How would I interpret $\widehat{\beta}_3 = 0.0037$?

The average change in bill length associated with an increase of 1g in body mass, holding species constant.

Note that we assume the slope is the same for each species!

Predictions

$$\begin{aligned}\widehat{\text{bill length}} = 24.92 &+ 9.92\ \text{IsChinstrap} + 3.56\ \text{IsGentoo} \\ &+ 0.0037\ \text{body mass}\end{aligned}$$

What is the predicted bill length for a Gentoo penguin with a body mass of 5000 g?

Predictions

$$\begin{aligned}\widehat{\text{bill length}} = 24.92 + 9.92\ \text{IsChinstrap} + 3.56\ \text{IsGentoo} \\ + 0.0037\ \text{body mass}\end{aligned}$$

> What is the predicted bill length for a Gentoo penguin with a body mass of 5000 g?

$24.92 + 3.56 + 0.0037(5000) = 46.98$

Fitting the model in R

```
m1 <- lm(bill_length_mm ~ species + body_mass_g,
                  data = penguins)
```

- `bill_length_mm ~ species + body_mass_g` is a `formula` in R
- Still `response ~ predictor(s)`
- `+` means "include more variables in the model"

Fitting the model in R

```
m1 <- lm(bill_length_mm ~ species + body_mass_g,
                 data = penguins)
summary(m1)
```

```
...
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)      2.492e+01  1.063e+00  23.443  < 2e-16 *
## speciesChinstrap 9.921e+00  3.511e-01  28.258  < 2e-16 *
## speciesGentoo    3.558e+00  4.858e-01   7.324 1.78e-12 *
## body_mass_g      3.749e-03  2.823e-04  13.276  < 2e-16 *
...
```

$$\begin{aligned}\widehat{\text{bill length}} = 24.92 &+ 9.92 \text{ IsChinstrap} + 3.56 \text{ IsGentoo} \\ &+ 0.0037 \text{ body mass}\end{aligned}$$

Class activity

Work on the class activity (handout).

Class activity

What is the population regression model?

Class activity

> What is the population regression model?

$$\text{Weight} = \beta_0 + \beta_1 \text{IsEnlarged} + \beta_2 \text{IsReduced} + \beta_3 \text{WingLength} + \varepsilon$$

Class activity

$$\widehat{\text{Weight}} = 1.94 - 0.46\text{IsEnlarged} + 0.90\text{IsReduced} + 0.45\text{WingLength}$$

> What is the average change in weight associated with a 1mm increase in wing length, for sparrows from enlarged nests?

Class activity

$$\begin{aligned}\widehat{\text{Weight}} = 1.94 - 0.46\text{IsEnlarged} + 0.90\text{IsReduced} + \\ 0.45\text{WingLength}\end{aligned}$$

> What is the average change in weight associated with a 1mm increase in wing length, for sparrows from enlarged nests?

Answer: 0.45

(We fit the same slope for each group of sparrows, and it is estimated to be 0.45)

Class activity

$$\widehat{\text{Weight}} = 1.94 - 0.46\text{IsEnlarged} + 0.90\text{IsReduced} + 0.45\text{WingLength}$$

What is the estimated difference in weight for sparrows from enlarged vs. control nests, with the same wing length?

Class activity

$$\widehat{\text{Weight}} = 1.94 - 0.46\text{IsEnlarged} + 0.90\text{IsReduced} + \\ 0.45\text{WingLength}$$

> What is the estimated difference in weight for sparrows from enlarged vs. control nests, with the same wing length?

We estimate that sparrows from enlarged nests weigh 0.46g less, on average, than sparrows from control nests with the same wing length